more about four-dimensional geometry than Einstein. Yet, in spite of that, Einstein did the work [formulated the general relativistic laws of gravity] and not the mathematicians." He did the work because mathematics was not enough; Einstein's unique physical insight was also needed.

Actually, Hilbert exaggerated. Einstein was a rather good mathematician, though in mathematical technique he was not the towering figure that he was in physical insight. As a result, few of Einstein's post-1912 arguments are presented today in the way Einstein presented them. People have learned improvements. And, with the quest to understand the laws of physics becoming more and more mathematical as the years after 1915 passed, Einstein became less and less the dominant figure he had been. The torch was passed to others.

---

1. See the note to page 61 for details.

2. It was not completely obvious that Newton's gravitational law violated Einstein's principle of relativity, because Einstein, in formulating his principle, had relied on the concept of an inertial reference frame, and this concept could not be used in the presence of gravity. (There is no way to shield a reference frame from gravity and thereby permit it to move solely under the influence of its own inertia.) However, Einstein was convinced that there must be some way to extend the sway of his relativity principle into the realm of gravity (some way to "generalize" it to include gravitational effects), and he was convinced that Newton's gravitational law would violate that yet-to-be-formulated "generalized principle of relativity."

3. Chapter 4 and especially Box 4.1

4. Einstein used the new phrase "general covariance" for this property, although it was just a natural extension of his principle of relativity